

AI TRADING RESEARCH LAB

ATRL — AI-assisted research, automation and experimental validation

An engineering research environment for repeatable trading-system experiments.

- 3 independent research bots with isolated experiment state
- Trade → Backtest → Lookahead → Hyperopt → A/B research
- 30-day primary validation windows with 7/14-day reliability checks
- Checkpoints, resume/recovery, diagnostics and run-state tracking
- Evidence-driven candidate selection instead of single-run optimization

Research loop

01 CONTROL

02 RUN

03 MEASURE

04 COMPARE

05 VALIDATE

Public portfolio separates engineering evidence from proprietary strategy implementation.

What the system does

The screenshots show a controlled research environment rather than a single trading script.

ORCHESTRATE

Start, inspect and coordinate multiple independent research stages.

EXPERIMENT

Run repeatable backtests, lookahead checks, hyperparameter searches and A/B candidates.

VALIDATE

Compare base vs candidate runs using trades, profit, PF, drawdown and hold duration.

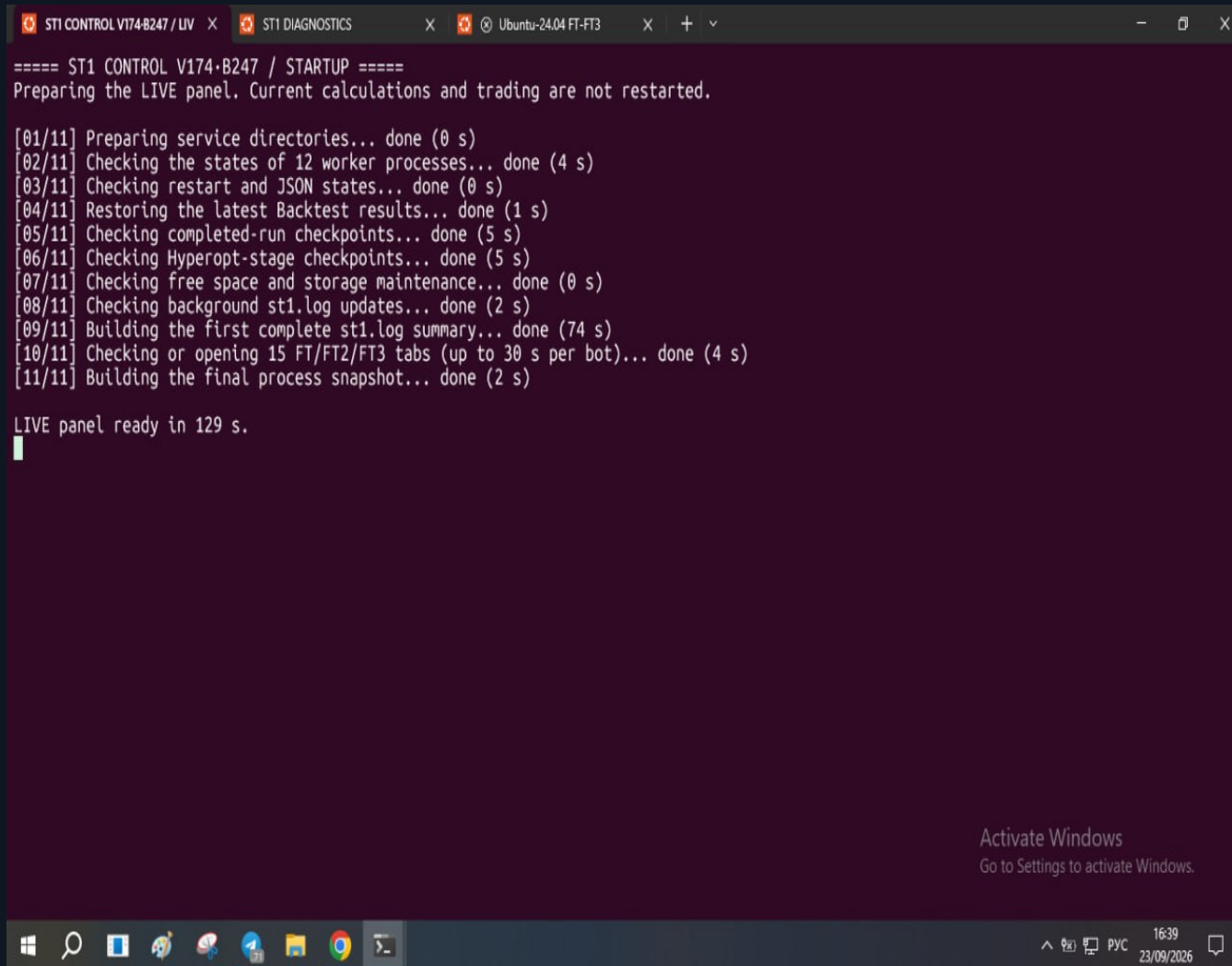
RECOVER

Persist state, checkpoints and diagnostics so long-running research can resume.

Observed UI vocabulary: RUNNING / DONE / SEARCH / REJECTED / BASE / PROSPECT / KEEP

Startup

11-step startup prepares the live panel without restarting current calculations or trading.



```
==== ST1 CONTROL V174-B247 / STARTUP ====
Preparing the LIVE panel. Current calculations and trading are not restarted.

[01/11] Preparing service directories... done (0 s)
[02/11] Checking the states of 12 worker processes... done (4 s)
[03/11] Checking restart and JSON states... done (0 s)
[04/11] Restoring the latest Backtest results... done (1 s)
[05/11] Checking completed-run checkpoints... done (5 s)
[06/11] Checking Hyperopt-stage checkpoints... done (5 s)
[07/11] Checking free space and storage maintenance... done (0 s)
[08/11] Checking background st1.log updates... done (2 s)
[09/11] Building the first complete st1.log summary... done (74 s)
[10/11] Checking or opening 15 FT/FT2/FT3 tabs (up to 30 s per bot)... done (4 s)
[11/11] Building the final process snapshot... done (2 s)

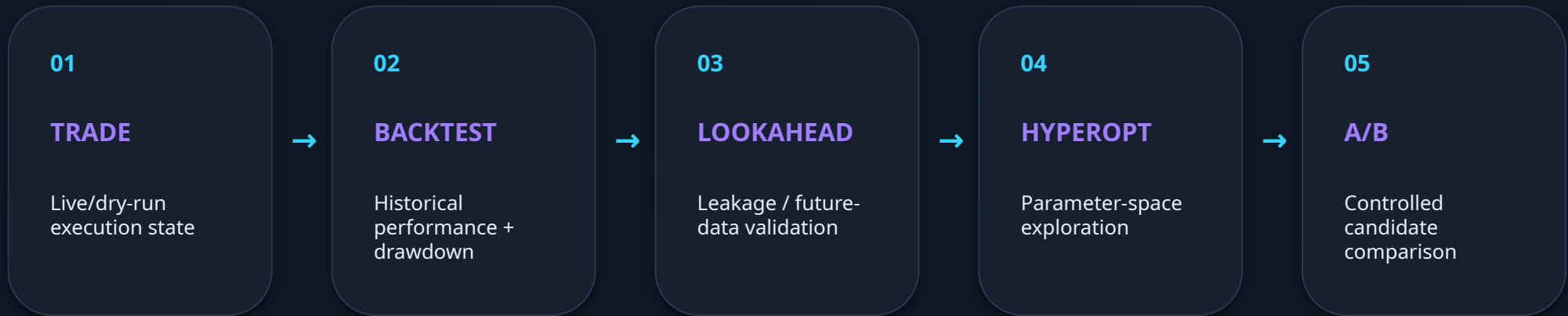
LIVE panel ready in 129 s.
```

What is visible

- Worker-process state checks
- Restart / JSON state checks
- Backtest and Hyperopt checkpoint restoration
- Storage / log maintenance
- Final process snapshot

Research execution pipeline

Each stage produces evidence consumed by the next stage.



Why this matters for QA / ML engineering

- The system separates generation of candidates from acceptance of candidates.
- Validation uses multiple windows and explicit baseline-vs-candidate evidence.
- Long-running research is treated as a stateful workflow with checkpoints and recovery.

ST1 Control — English UI

Live overview for FT, FT2 and FT3; one control plane for research state.

```
ST1 CONTROL V174-B247 / LIV x ST1 DIAGNOSTICS x ST1 A/B TESTS x ALL HYPEROPT x + v - [ ] X

===== ST1 CONTROL V174-B247 / Updated: 2026-09-25 16:08:58 | FT=30d/V7/300c | FT2=30d/V7/800c | FT3=30d/V7/350c =====
===== OVERVIEW =====
BOT | TRADE | LOOK | BACKTEST | HYPEROPT
-----
FT | RUNNING | DONE | 14 tr. + +7.768 USDT + DD 0.499% | BUY 1/10 + 393/800 + cycle 8 + repeat 1/2
FT2 | RUNNING | DONE | 14 tr. + +8.673 USDT + DD 1.719% | SLCORE 9/15 + 179/700 + cycle 4 + repeat 1/1
FT3 | RUNNING | DONE | 27 tr. + +7.735 USDT + DD 3.035% | SLCORE 7/13 + 504/700 + cycle 3 + repeat 2/2

===== OPTIMIZATION | HYP LEADER → LAST CAND | trades / profit USDT =====
BOT | [UNTRANSLAT] CYCLE | STAGE | HYP [UNTRA] | LAST CAND | RESULT
-----
FT | CAROUSEL | 8 | BUY + repeat 1 | 13/+5.368 | 9/+3.979 | REJECTED
FT2 | CAROUSEL | 4 | SLCORE + repeat 1 | 14/+6.232 | 14/+5.861 | REJECTED
FT3 | CAROUSEL | 3 | SLCORE + repeat 2 | 29/+4.924 | 29/+4.924 | ACCEPTED

===== DECISIONS | LAST CAND → MAIN → reliable 7D | 1H/6H/1D diagnostic =====
FT | REJECTED | FINAL: MAIN: risk-adjusted profit +0.036 < floor +0.039
FT2 | REJECTED | SCORE: MAIN: no material improvement
FT3 | ACCEPTED | SLCORE: MAIN improved + reliable 7D confirmed + ΔP +0.014

===== A/B TESTS B283 PHASE-ISOLATED | ENGINE R4/B283 | EACH BOT INDEPENDENT | SENS 6x5D → BTC30 → RELIABLE 7/14D =====
BOT | STAGE | CANDIDATE | DECISION | 30D BASE-TEST | 7/14D BASE-TEST | P30
-----
FT | R9C23/0/4 | A_CURRENT → B_R9_FT_EXI_ROLLOVER | SEARCH | BTC30 0/4 | NEXT 7/14D | OFF
FT2 | R9C16/9/1 | B_R9_FT2_ENT_RECLAIM080 → B_R9_FT2_ENT_SIG | SEARCH | BTC30 7/12 | NEXT 7/14D | OFF
FT3 | R9C14/4/7 | B_R9_FT3_FAL_MAXDUR84 → B_R9_FT3_FAL_MAXDU | SEARCH | BTC30 3/7 | NEXT 7/14D | OFF

===== SETTINGS =====
[Enter/1] All 3 bots with defaults [2] Settings for all 3 ↗ [4/5/6] Manage FT / FT2 / FT3 ↗
[3] Deep hyperopt ↗ [8] Hyperopt carousel: FT+FT2+FT3 ↗ [C] Resume Hyperopt ↗
[7] Three Hyperopt LIVE ↗ [9] Defaults + carousel ↗ [A] Diagnostics ↗ + [H/?] Help ↗
[B] A/B tests ↗ [G] Git: project + diagnostics + trades ↗ [S] Global stop ↗
[I] Impact/restart ↗ [L] Language: ENG ↔ RUS [0] Close ST1
```

Main functions

- Trade / Lookahead / Backtest / Hyperopt state
- Hyperopt cycle, stage, repeat and latest candidate
- 30D base→test and 7/14D reliability checks
- A/B candidate search and decision state
- Diagnostics, Help, Git, Restart and Language

A/B Tests — isolated candidate research

Independent candidate comparison with baseline evidence and explicit decisions.

ST1 CONTROL V174-B247 / LIVI

ST1 DIAGNOSTICS

ST1 A/B TESTS

ALL HYPEROPT

===== A/B TESTS B283 PHASE-ISOLATED | ENGINE R4/B283 | EACH BOT INDEPENDENT | SENS 6x5D → BTC30 → RELIABLE 7/14D =====

BOT	STAGE	CANDIDATE	DECISION	30D BASE-TEST	7/14D BASE-TEST	P30
FT	R9C23/0/4	A_CURRENT → B_R9_FT_EXI_ROLLOVER	SEARCH	BTC30 0/4	NEXT 7/14D	OFF
FT2	R9C16/9/1	B_R9_FT2_ENT_RECLAIM080 → B_R9_FT2_ENT_SIG	SEARCH	BTC30 7/12	NEXT 7/14D	OFF
FT3	R9C14/5/7	B_R9_FT3_FAL_MAXDUR96 → B_R9_FT3_FAL_MAXDU	SEARCH	BTC30 4/7	NEXT 7/14D	OFF

1) All 3 (R9C23/0/4/R9C16/9/12/R9C14/5/7)

4) FT3 (R9C14/5/7)

7) Stop A/B

2) FT (R9C23/0/4)

5) Priority

[Space] Update

3) FT2 (R9C16/9/12)

6) Reasons + paths

0) Back

===== A/B TEST - CURRENT RUN =====

BOT	STEP	RUN	TEST	TR	PROFIT	PF	DD	HOLD	RESULT
FT3	R4/BASE	20260925	A_CURRENT	27	+4.767	1.34	3.1%	13h19m	BASE
FT	R4/BASE	20260925	A_CURRENT	14	+5.803	3.22	1.2%	3h56m	BASE
FT	R4/4/8	20260925	B_R9_FT_FAL_BE420	14	+5.281	3.02	1.3%	4h05m	REJECT
FT	R4/7/8	20260925	B_R9_FT_FAL_MINP003	14	+5.213	2.99	1.3%	3h50m	REJECT
FT	R4/3/8	20260925	B_R9_FT_FAL_BE356	14	+5.045	2.93	1.3%	3h58m	REJECT
FT	R4/2/8	20260925	B_R9_FT_FAL_BE300	14	+5.029	2.92	1.3%	3h54m	REJECT
FT3	R4/BASE	20260925	A_CURRENT	14	+3.302	1.25	4.0%	1d11h	BASE
FT3	R4/3/4	20260925	B_R9_FT3_EXI_ROLLOVER_RSI68	22	+0.312	1.02	3.1%	19h31m	REJECT
FT	R4/BASE	20260925	A_CURRENT	14	+5.803	3.22	1.2%	3h56m	BASE
FT2	R4/BASE	20260925	A_CURRENT	13	+6.436	3.59	1.8%	8h15m	BASE
FT2	R4/3/11	20260925	B_R9_FT2_FAL_MAXDUR60	13	+6.724	3.71	1.8%	7h33m	PROMISING
FT2	R4/5/11	20260925	B_R9_FT2_FAL_MAXDUR96	13	+6.324	3.55	1.8%	8h27m	REJECT
FT2	R4/4/11	20260925	B_R9_FT2_FAL_MAXDUR84	13	+6.252	3.52	1.8%	8h13m	REJECT
FT2	R4/2/11	20260925	B_R9_FT2_FAL_MAXDUR44	13	+4.574	2.84	1.8%	6h37m	REJECT
FT	R4/BASE	20260925	A_CURRENT	14	+4.738	2.81	1.3%	4h15m	BASE
FT	R4/BASE	20260925	A_CURRENT	14	+5.803	3.22	1.2%	3h56m	BASE

Evidence shown

- FT / FT2 / FT3 independently
- BASE and candidate runs
- Trades, profit, PF, DD and hold time
- KEEP / PROSPECT / REJECT / BASE outcomes
- Run and step traceability

FT3 Backtest

Historical Freqtrade result used as one validation layer.

[illegible]

Observed output

- Backtest result table and summary
- Trade-level and aggregate metrics
- Profit / drawdown / duration fields
- Concrete historical validation evidence

Hyperopt: all three bots

Parallel Hyperopt progress for FT, FT2 and FT3.

ST1 CONTROL V174-B247 / X ST1 DIAGNOSTICS X ST1 A/B TESTS X ST1 DEEP HYPEROPT X ALL HYPEROPT X + -									
Best	278/800	5	10 44.4 4 0	0.10%	(0.93%) 0.487 USDT	3:42:00	1.93945	(2.44%) 0.345 USDT	
Best	298/800	6	1 80.0 3 0	0.62%	(0.44%) 3.735 USDT	3:18:00	1.79784	(0.31%) 0.506 USDT	
Best	309/800	18	3 50.0 8 0	0.25%	(3.36%) 4.472 USDT	3:30:00	-1.92378	(0.46%) 2.621 USDT	
Best	390/800	8	10 44.4 5 0 3 62.5	0.55%	(4.03%) 4.376 USDT (3.94%)	4:21:00	-2.33474	(2.23%) 0.532 USDT (0.48%)	
Epochs 391/800 49% • 2:54:30 • 2:22:17									
Best	73/700	9	3 66.7 6 0	0.42%	(3.38%) 3.757 USDT	10:02:00	2.47069	(1.74%) 1.997 USDT	
Best	127/700	9	3 66.7 6 0	0.42%	(3.38%) 3.830 USDT	10:02:00	2.18549	(1.73%) 1.923 USDT	
Best	138/700	9	3 66.7 6 0	0.43%	(3.45%) 3.833 USDT	10:02:00	2.17836	(1.66%) 1.921 USDT	
Best	173/700	9	3 66.7 6 0 3 66.7	0.44%	(3.45%) 3.995 USDT (3.60%)	9:58:00	2.02787	(1.66%) 1.923 USDT (1.66%)	
Epochs 178/700 25% • 2:53:36 • -:--:--									
Best	287/700	20	9 52.6 11 0	0.22%	(3.94%) 4.447 USDT	11:20:00	9.49367	(1.80%) 2.175 USDT	
Best	296/700	20	9 55.0 11 0	0.22%	(4.01%) 4.475 USDT	11:20:00	9.38016	(1.85%) 2.136 USDT	
Best	352/700	20	9 55.0 11 0	0.22%	(4.03%) 4.487 USDT	11:21:00	9.32663	(1.82%) 2.115 USDT	
Best	426/700	20	9 55.0 11 0	0.22%	(4.04%) 4.497 USDT	11:20:00	9.27471	(1.80%) 2.094 USDT	

Research visibility

- Separate epoch streams
- Best candidates update during the run
- Trades, profit, objective and drawdown
- Long-running optimization remains observable

Defaults + Hyperopt Carousel

Select FT, FT2, FT3 or all three and restart research from bot defaults.

```
ST1 CONTROL V174-B247 / X ST1 DIAGNOSTICS X ST1 A/B TESTS X ALL HYPEROPT X ST1 DEFAULTS + CAROUSEL X + - [icon] X

===== DEFAULTS + HYPEROPT CAROUSEL / SELECT BOTS =====

[Enter/1] All 3 bots: FT + FT2 + FT3
[2] FT only
[3] FT2 only
[4] FT3 only
[0] Close tab

For selected bots:
TRADE + BACKTEST + LOOKAHEAD restart with each bot's defaults.
HYPEROPT restarts only as a fresh default carousel.
Existing ST1 state/raw/checkpoint/carousel logs remain enabled.
Choice [1]: █
```

Operational control

- All 3 bots can be selected
- Trade + Backtest + Lookahead restart
- Hyperopt starts as a fresh default carousel
- Existing state / checkpoint logs remain enabled

FT Control

Per-bot lifecycle control and current process state.

```
ST1 CONTROL V174-B247 / X ST1 DIAGNOSTICS X ST1 A/B TESTS X ALL HYPEROPT X FT CONTROL X + -
```

```
===== FT / CONTROL =====
```

PROCESS	STATUS	TIME	EXI	ERROR	PHASE
ft_trade	RUNNING	1-08:52:58	-	0	dry-run trading is running
ft_backtest	DONE	-	0	0	2880 c 14 tr. +7.768 USDT +6.998% DD 0.499% avg. 4:12
ft_lookahead	DONE	-	0	0	90d no lookahead bias
ft_hyperopt	RUNNING	03:05:36	-	0	CAR/ProfitTrades S1/2 BUY 1/10 402/800 2208 c R8

```
Defaults FT: period=30d | holdout=7d | warmup=300c | tf=15m
```

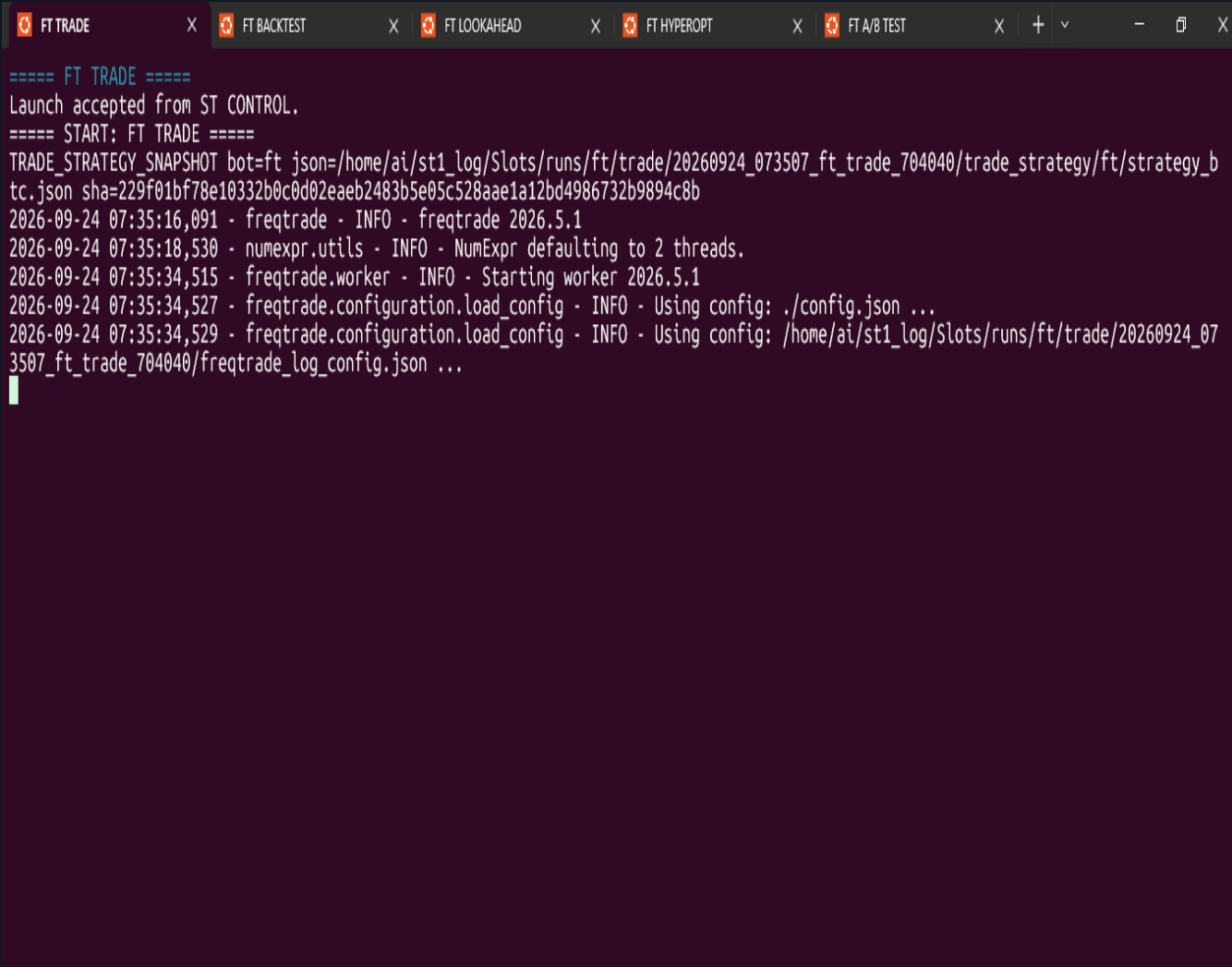
```
1) Restart TRADE
2) Start/restart BACKTEST
3) Start/restart LOOKAHEAD
4) HYPEROPT
5 / Enter) Restart the selected bot with defaults
6) Stop a mode
7) Configure and restart the entire bot
0) Back
Selection [5]:
```

Available actions

- Restart Trade
- Start / restart Backtest
- Start / restart Lookahead
- Hyperopt
- Stop a mode
- Configure and restart the bot

FT Trade

Real Freqtrade startup evidence from the FT Trade window.

A screenshot of a terminal window titled "FT TRADE" with several other tabs open: "FT BACKTEST", "FT LOOKAHEAD", "FT HYPEROPT", and "FT A/B TEST". The terminal displays the following text:

```
==== FT TRADE ====
Launch accepted from ST CONTROL.
==== START: FT TRADE ====
TRADE_STRATEGY_SNAPSHOT bot=ft json=/home/ai/st1_log/Slots/runs/ft/trade/20260924_073507_ft_trade_704040/trade_strategy/ft/strategy_b
tc.json sha=229f01bf78e10332b0c0d02eae2483b5e05c528aae1a12bd4986732b9894c8b
2026-09-24 07:35:16,091 - freqtrade - INFO - freqtrade 2026.5.1
2026-09-24 07:35:18,530 - numexpr.utils - INFO - NumExpr defaulting to 2 threads.
2026-09-24 07:35:34,515 - freqtrade.worker - INFO - Starting worker 2026.5.1
2026-09-24 07:35:34,527 - freqtrade.configuration.load_config - INFO - Using config: ./config.json ...
2026-09-24 07:35:34,529 - freqtrade.configuration.load_config - INFO - Using config: /home/ai/st1_log/Slots/runs/ft/trade/20260924_07
3507_ft_trade_704040/freqtrade_log_config.json ...
```

Visible evidence

- ST Control launch accepted
- Strategy snapshot captured
- Freqtrade version shown
- Config and runtime log loaded

FT2 Lookahead

Lookahead Analysis result and bias check.

```
FT2 TRADE X FT2 BACKTEST X FT2 LOOKAHEAD X FT2 HYPEROPT X FT2 A/B TEST X + - [ ] X

2026-09-23 20:34:49,222 - freqtrade.strategy.hyper - INFO - Strategy Parameter: sl_trail_atr_mult = 3.63
2026-09-23 20:34:49,223 - freqtrade.strategy.hyper - INFO - Strategy Parameter: sl_trail_start_profit = 0.005
2026-09-23 20:34:49,584 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-09-23 00:00:00 (98 days).
2026-09-23 20:36:57,386 - freqtrade.loggers.set_log_levels - INFO - Reducing verbosity for bias tester.
2026-09-23 20:36:57,389 - freqtrade.optimize.analysis.lookahead - INFO - Found 57 trades, calculating 20 trades.
2026-09-23 20:36:57,611 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-26 02:15:00 (9 days).
2026-09-23 20:37:06,342 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-26 02:15:00 (9 days).
2026-09-23 20:37:16,430 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-26 21:30:00 (10 days).
2026-09-23 20:37:27,345 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-27 14:45:00 (10 days).
2026-09-23 20:37:40,860 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-28 11:15:00 (11 days).
2026-09-23 20:37:53,634 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-28 13:15:00 (11 days).
2026-09-23 20:38:07,406 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 00:15:00 (13 days).
2026-09-23 20:38:22,387 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 00:15:00 (13 days).
2026-09-23 20:38:39,542 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 08:15:00 (13 days).
2026-09-23 20:38:52,167 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 09:30:00 (13 days).
2026-09-23 20:39:08,091 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 09:30:00 (13 days).
2026-09-23 20:39:24,336 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 11:30:00 (13 days).
2026-09-23 20:39:40,410 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 13:45:00 (13 days).
2026-09-23 20:39:54,592 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 13:45:00 (13 days).
2026-09-23 20:40:11,850 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-06-30 23:30:00 (14 days).
2026-09-23 20:40:28,561 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-01 00:30:00 (14 days).
2026-09-23 20:40:44,876 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-02 07:00:00 (15 days).
2026-09-23 20:41:03,288 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-02 07:00:00 (15 days).
2026-09-23 20:41:23,340 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-05 17:15:00 (19 days).
2026-09-23 20:41:46,359 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-05 23:45:00 (19 days).
2026-09-23 20:42:14,382 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-07 04:45:00 (20 days).
2026-09-23 20:42:43,395 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-07 05:00:00 (20 days).
2026-09-23 20:43:10,790 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-09 03:45:00 (22 days).
2026-09-23 20:43:41,456 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-09 07:30:00 (22 days).
2026-09-23 20:44:16,880 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-10 18:45:00 (24 days).
2026-09-23 20:44:56,547 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-10 19:00:00 (24 days).
2026-09-23 20:45:37,411 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-13 19:00:00 (27 days).
2026-09-23 20:46:16,805 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-14 01:30:00 (27 days).
2026-09-23 20:46:55,407 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-18 00:00:00 (31 days).
2026-09-23 20:47:38,961 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-18 00:15:00 (31 days).
2026-09-23 20:48:23,422 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-20 23:30:00 (34 days).
2026-09-23 20:49:10,690 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-21 08:45:00 (34 days).
2026-09-23 20:49:59,299 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-21 16:00:00 (37 days).
2026-09-23 20:50:53,437 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-23 16:45:00 (37 days).
2026-09-23 20:51:51,434 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-23 17:30:00 (37 days).
2026-09-23 20:52:45,681 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-23 18:00:00 (37 days).
2026-09-23 20:53:44,587 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-23 18:45:00 (37 days).
2026-09-23 20:54:40,601 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-23 20:00:00 (37 days).
2026-09-23 20:55:44,895 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-24 11:45:00 (37 days).
2026-09-23 20:56:51,655 - freqtrade.optimize.backtesting - INFO - Loading data from 2026-06-16 16:00:00 up to 2026-07-24 12:00:00 (37 days).
2026-09-23 20:57:47,658 - freqtrade.optimize.analysis.lookahead - INFO - Found targeted trade amount = 20 signals.
2026-09-23 20:57:47,669 - freqtrade.loggers.set_log_levels - INFO - Restoring log verbosity.
2026-09-23 20:57:47,671 - freqtrade.optimize.analysis.lookahead - INFO - strategy_bt2: no bias detected
2026-09-23 20:57:47,684 - freqtrade.optimize.analysis.lookahead_helpers - INFO - Checking look ahead bias via backtests of strategy_bt2.py took 1384 seconds.
Lookahead Analysis

filename strategy has_bias total_signals biased_entry_signals biased_exit_signals biased_indicators
strategy_bt2.py strategy_bt2 No 20 0 0 0

2026-09-23 20:57:47,800 - freqtrade.optimize.analysis.lookahead_helpers - INFO - saving /home/ai/stl_log/Slots/runs/ft2/lookahead/20260923_203420_ft2_lookahead_119970/ft2_lookahead.csv
LOOKAHEAD OK: CSV has_bias=false; biased_entry=0; biased_exit=0

==== DONE / STOPPED: FT2 LOOKAHEAD / exit=0 ====

COMMAND WITH LIVE STATUS:
bash /tmp/stl_slots/generated/ft2/lookahead/20260923_203420_ft2_lookahead_119970/ft2_lookahead.sh

==== DONE: FT2 LOOKAHEAD / exit=0 ====
[Enter/R] repeat with defaults | [S] stop | [L] open log
```

Validation purpose

- Historical data processed
- Lookahead analysis executed
- Bias result is explicit
- CSV evidence is saved

FT Hyperopt

Live Hyperopt result table for the FT research stream.

```
FT TRADE X FT BACKTEST X FT LOOKAHEAD X FT HYPEROPT X FT A/B TEST X + - X
```

```
/home/ai/st1_log/Slots/runs/ft/hyp/20260925_131249_ft_hyp_2861024/freqtrade_log_config.json ...  
/home/ai/ft/.env/lib/python3.12/site-packages/freqtrade/optimize/hyperopt/hyperopt_optimizer.py:435: FutureWarning: `optuna.terminator`  
module has been deprecated in v4.9.0. This feature will be removed in v6.0.0. See https://github.com/optuna/optuna/releases/tag/v4  
.9.0.  
self.es_terminator = Terminator(BestValueStagnationEvaluator(self.es_epochs))  
[I 2026-09-25 13:16:05,273] A new study created in memory with name: no-name-9870b194-783d-4d5c-867b-aabdbd5a3397  
Hyperopt results
```

Best	Epoch	Trades	Win Loss	Draw Win%	Avg profit	Profit	Avg duration	Objective	Max Drawdown (Acct)
* Best	9/800	8	2	0	-0.40%	-3.186 USDT (-2.87%)	3:02:00	48.40621	3.524 USDT (3.17%)
* Best	30/800	6	0	0	-0.46%	-2.735 USDT (-2.46%)	2:40:00	47.68963	2.735 USDT (2.46%)
Best	64/800	6	2	0	-0.20%	-1.207 USDT (-1.09%)	5:28:00	25.05221	1.322 USDT (1.19%)
Best	71/800	10	4	0	0.03%	0.342 USDT (0.31%)	4:27:00	9.07953	3.110 USDT (2.72%)
Best	81/800	7	4	0	-0.00%	-0.026 USDT (-0.02%)	4:09:00	6.73367	0.456 USDT (0.41%)
Best	166/800	18	8	0	0.06%	1.031 USDT (0.93%)	4:30:00	4.24700	2.803 USDT (2.44%)
Best	278/800	5	10	44.4	0.10%	0.487 USDT (0.44%)	3:42:00	1.93945	0.345 USDT (0.31%)
Best	298/800	6	1	80.0	0.62%	3.735 USDT (3.36%)	3:18:00	1.79784	0.506 USDT (0.46%)
Best	309/800	18	3	50.0	0.25%	4.472 USDT (4.03%)	3:30:00	-1.92378	2.621 USDT (2.23%)
Best	390/800	8	8	0	0.55%	4.376 USDT (3.94%)	4:21:00	-2.33474	0.532 USDT (0.48%)
Best	398/800	16	3	62.5	0.34%	5.474 USDT	3:56:00	-4.49468	1.044 USDT

Visible metrics

- Epoch and best-so-far
- Trades and win/loss
- Average profit and total profit
- Objective and max drawdown

Resume Hyperopt

Saved Hyperopt series can be inspected and resumed after preflight.

ST1 CONTROL V174 X ST1 DIAGNOSTICS X ST1 A/B TESTS X ALL HYPEROPT X FT CONTROL X ST1 RESUME HYPEROPT X

===== RESUME HYPEROPT =====

#	BOT	STARTED	HYPEROPT	DONE	NEXT STAGE	TRADES	PROFIT	AVG.TIME	STATUS
1	FT	24.09 18:46	CAR/MultiMetric	8/10	STOPLOSS_EMERGENCY	12	+5.964 USDT	3:42	READY
2	FT	24.09 07:39	CAR/MultiMetric	8/10	STOPLOSS_EMERGENCY	10	+6.210 USDT	3:39	READY
3	FT2	25.09 01:52	CAR/StopRiskV3	8/15	SLCORE_S1	10	+1.968 USDT	6:34	ACTIVE
4	FT2	23.09 20:37	CAR/MultiMetric	10/15	STOPLOSS_EMERGENCY	9	+3.822 USDT	10:02	READY
5	FT3	24.09 05:00	CAR/StopRiskV3	6/13	SLCORE_S2	20	+5.162 USDT	11:11	ACTIVE

Blocked by safe preflight: 39 series. [D] Show reasons.

↑/↓ - select | Enter - details and launch | R - refresh | 0 - close

0/N series are hidden; inactive 0/N series are safely archived in /home/ai/_bak and removed.

Recovery capability

- Saved series and next stage
- Trades / profit / average time
- READY / ACTIVE state
- Safe preflight can block unsafe series

ST1 Control — Russian UI

The same live control plane in Russian; demonstrates ENG ↔ RUS localization.

```
ST1 DIAGNOSTICS | ST1 A/B TESTS | Ubuntu-24.04 FT-FT3
ИЕ V174-B247 / Обновлено: 2026-09-20 16:28:42 | FT=30д/К7/300св | FT2=30д/К7/800св | FT3=30д/К7/350св
=====
| LOOK | BACKTEST | HYPEROPT |
=====
| ГОТОВО | 29 сдел. · +10.840 USDT · DD 3.445% | SLCORE 6/10 · 422/700 · круг 1 · повтор 1/2 |
| ГОТОВО | 28 сдел. · +2.920 USDT · DD 3.614% | REGIME 6/15 · 103/500 · круг 1 · повтор 1/2 |
| ГОТОВО | 32 сдел. · +7.623 USDT · DD 7.287% | EXITCB 5/13 · 116/500 · круг 1 · повтор 1/2 |
=====
| НУР ЛИДЕР → ПОСЛ. КАНД | сделки / прибыль USDT | =====
| КРУГ | ЭТАП | | НУР ЛИДЕР | ПОСЛ. КАНД | ИТОГ |
=====
| 1 | SLCORE · повтор 1 | 29/+5.869 | 29/+5.869 | ПРИНЯТ |
| 1 | REGIME · повтор 1 | 31/+1.952 | 31/+1.166 | ОТКЛОНЁН |
| 1 | EXITCB · повтор 1 | 32/-1.323 | 35/-2.079 | ОТКЛОНЁН |
=====
ПОСЛ. КАНД → MAIN → надёжный 7д | 14/64/1д — диагностика =====
EXITCB: MAIN улучшен · надёжный 7д подтвердил · ΔР +0.002
EXITCB: MAIN: риск-прибыль +0.011 < порог +0.013
SELL: MAIN: прибыль хуже базы
=====
283 PHASE-ISOLATED | ENGINE R4/B283 | КАЖДЫЙ БОТ НЕЗАВИСИМ | SENS 6×5д → BTC30 → НАДЁЖНЫЙ 7/14д =====
КАНДИДАТ | РЕШЕНИЕ | 30д БАЗА-КАНД | 7/14д БАЗА-КАНД |
=====
B_R9_FT_EXI_HYBRID → B_R9_FT_EXI_ROLLOVER | ПОИСК | BTC30 1/4 | NEXT 7/14D |
B_R9_FT2_ENT_SIGNAL3 → B_R9_FT2_ENT_TRIGGER | ПОИСК | BTC30 1/12 | NEXT 7/14D |
B_R9_FT3_EXI_RSI_PEAK_ROLLOVER → B_R9_FT3_ | ПОИСК | BTC30 0/4 | NEXT 7/14D |
=====
[2] Настройки всех 3 ↗ | [4/5/6] Управление FT / FT2 |
[8] Карусель Нурегорт: FT+FT2+FT3 ↗ | [C] Продолжить Нурегорт ↗ |
[9] Дефолты + Карусель ↗ | [A] Диагностика ↗ · [H/?] По |
[G] Git: проект + диагностика + сделки ↗ | [S] Глобальная остановка ↗ |
[L] Language: ENG → RUS | [0] Закреть ST1 Activate Win |
Go to Settings to |
```

User-facing support

- Same FT / FT2 / FT3 control model
- Russian operational labels
- Diagnostics / Help / Git / A/B controls
- Language switch is part of ST1

Three public-safe strategy profiles

15m LONG research strategies • high-level logic only • proprietary thresholds and scoring excluded

FT — strategy_btc

15m LONG

Extrema / rebound research

- EMA
- RSI
- StochRSI
- Bollinger Bands
- ATR

Trading logic — high level

Extrema/pivot context is combined with rebound and distance conditions. RSI, StochRSI and Bollinger provide confirmation; ATR supports protection / distance-aware logic.

No proprietary thresholds / weights / exact scoring.

FT2 — strategy_btc2

15m LONG

Multi-signal consensus

- RSI
- EMA
- ATR
- Bollinger Bands
- ZigZag / extrema

Trading logic — high level

Entry can combine extrema, RSI and EMA-stretch signals. Consensus controls independent signal agreement; reclaim paths use Bollinger, extrema distance and ATR gates.

No proprietary thresholds / weights / exact scoring.

FT3 — strategy_btc3

15m LONG

ZigZag + RSI

- RSI
- ATR

Trading logic — high level

ZigZag/extrema is a primary signal component. Entry combines RSI + extrema with ATR-aware gates; exits include peak / RSI-rollover style models.

No proprietary thresholds / weights / exact scoring.

QA / validation engineering

The research system applies QA principles to experimental ML/trading workflows.

REPRODUCIBILITY

Fixed periods, repeat/cycle tracking, explicit baselines and candidate identifiers.

REGRESSION CONTROL

Compare candidate outputs against a known baseline before progression.

OBSERVABILITY

Process snapshots, logs, diagnostics, stage state and measurable outcomes.

FAILURE ANALYSIS

Inspect rejected candidates, risk-adjusted thresholds and diagnostic windows.

RECOVERY

Checkpoints and resumable execution reduce loss of research progress.

EVIDENCE

Decisions are tied to measured results rather than a single optimization score.

Technology stack

Technologies directly reflected by the project workflow and public portfolio.

UBU

PLATFORM

Ubuntu / Linux
WSL + Windows host
CLI / terminal workflows

PY

ENGINEERING

Python
Bash
Git / GitHub
GitHub Actions / CI

FT

TRADING

Freqtrade
CCXT
Bybit dry-run
BTC/USDT research

NP

DATA / SCIENCE

Pandas
NumPy
SciPy
Peak / extrema analysis

SIG

INDICATORS / SIGNALS

EMA • RSI • Bollinger Bands
ATR • ZigZag
find_peaks • EMD

QA

QA / RESEARCH

Backtesting
Lookahead analysis
Hyperopt
A/B experiments
Holdout / reliability checks

Python / data / ML engineering

The portfolio emphasizes engineering around experiments, not only model or strategy code.

ORCHESTRATION

Python + Bash controllers • worker lifecycle • state machines • checkpoints

DATA PIPELINE

Pandas / NumPy • time-series transforms • feature and indicator computation

SIGNAL ANALYSIS

SciPy peak/extrema detection • ZigZag • EMD • EMA/RSI/BB/ATR-derived features

EXPERIMENT ENGINE

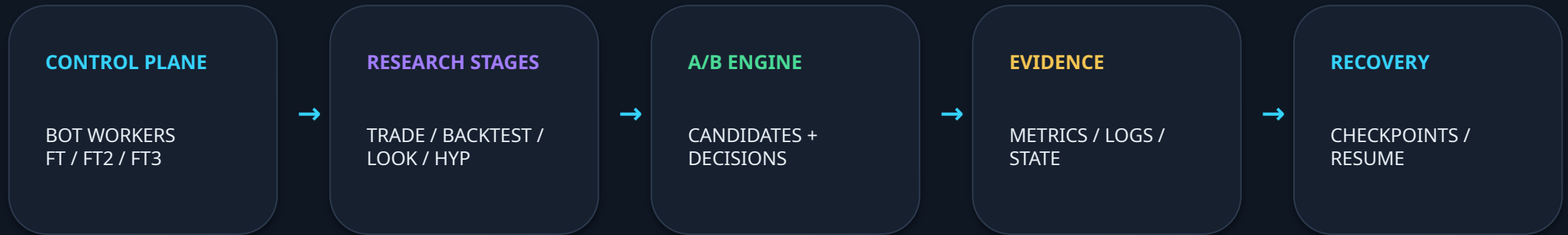
Hyperopt • deterministic runs • repeated candidates • isolated A/B tests

VALIDATION LAYER

Backtest • lookahead • holdout windows • drawdown / PF / duration metrics

System architecture

Conceptual architecture derived from the control workflow shown in the screenshots.



The key engineering property is separation of control, execution, experiment state and evidence.

Public repository exposes the engineering story; proprietary strategy implementation and credentials remain private.

Continuous optimization loop

Cyclic Hyperopt + A/B validation refreshes candidates through repeated research cycles.

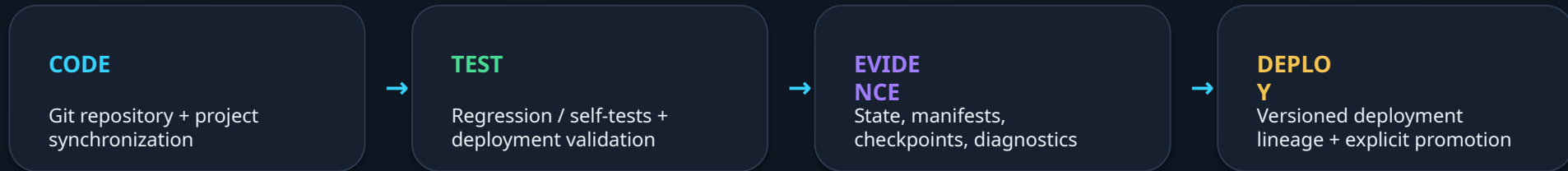


ADAPTIVE RESEARCH LOOP

The carousel repeatedly evaluates candidates, compares them with the current baseline and carries validated candidates into the next research cycle. It is an automated, continuously refreshed research workflow — not a claim of autonomous machine learning.

Git, validation & controlled deployment

Version control, validation evidence and deployment lineage are integrated into the engineering lifecycle.



Engineering controls

- Git-based project synchronization and versioned research state.
- Regression/self-tests and deployment validation are part of the workflow.
- Deployment lineage preserves previous and selected states with evidence.
- Research candidates are not deployed implicitly: promotion remains controlled.
- Checkpoints and manifests provide traceability for long-running experiments.

Public repository

Trd-cra / AI-Trading-Research-Lab

github.com/Trd-cra/AI-Trading-Research-Lab

Public: workflow, validation and selected evidence. Private: proprietary strategy implementation and credentials.

Contact & Portfolio

Senior QA Engineer | AI/ML QA & Automation | Web, API & Data

PUBLIC PORTFOLIO

AI Trading Research Lab (ATRL)

github.com/Trd-cra/AI-Trading-Research-Lab

LINKEDIN

Professional profile

linkedin.com/in/alex-k-46448175

Portfolio focus: AI-assisted QA • research automation • Hyperopt • A/B validation • reproducibility • diagnostics